$$E(X) = \int_{-\infty}^{\infty} xf(x)\,dx \qquad \text{provided the integral exists}$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)\,dx$$

NOTE

If $g(X)$ is a function of a random variable and $E[g(X)]$ exists then

Results:

(1) $E(c) = c$ where c is constant

Proof :

By definition, $E(X) = \int_{-\infty}^{\infty} xf(x)\,dx$

$$E(c) = \int_{-\infty}^{\infty} cf(x)\,dx = c\int_{-\infty}^{\infty} f(x)\,dx$$

$$= c \times 1 = c \qquad \left(\because \int_{-\infty}^{\infty} f(x)\,dx = 1\right)$$

(2) $E(aX) = a\,E(X)$

Proof :

$$E(aX) = \int_{-\infty}^{\infty} axf(x)\,dx = a\int_{-\infty}^{\infty} xf(x)\,dx$$

$$= a\,E(X)$$

Example 9.14

The p.d.f. of a continuous random variable X is given by

$$f(x) = \begin{cases} \frac{x}{2}, & 0 < x < 2 \\ 0, & \textit{elsewhere} \end{cases}$$ find its mean and variance

Solution:

$$E(X) = \int_{-\infty}^{\infty} xf(x)\,dx$$

$$E(X) = \int_{0}^{2} x\left(\frac{x}{2}\right)dx$$

$$= \frac{1}{2}\int_{0}^{2} x^2\,dx$$

$$= \frac{1}{2}\left[x^3/3\right]_0^2$$